PREDICTIVE GRID INTELLIGENCE FRAMEWORK FOR DISTRIBUTION SUBSTATION MAINTENANCE FAILURE PREVENTION THROUGH ADVANCED MACHINE LEARNING ANALYTICS

Project Phase-I Report submitted to Osmania University, Hyderabad,
in partial fulfillment of the requirement for the award of the degree of

BACHELOR OF ENGINEERING

in

INFORMATION TECHNOLOGY

By

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Under the Guidance of

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CERTIFICATE

This is to certify that the Project Phase-I report entitled "**Predictive Grid Intelligence Framework for Distribution Substation Maintenance Failure Prevention Through Advanced Machine Learning Analytics**" being submitted by **Mr. Mohammed Abdul Rahman (160923737111)** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering** in **Information Technology** is a record of bonafide work carried out by him.

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DECLARATION BY THE CANDIDATES

I, Mr. Mohammed Abdul Rahman (160923737111) hereby declare that the Project Phase-I report entitled "**Predictive Grid Intelligence Framework for Distribution Substation Maintenance Failure Prevention Through Advanced Machine Learning Analytics**" is carried out under the guidance of **Mr. M Shiva Rama Krishna**, Assistant Professor, Department of Information Technology, Lords Institute of Engineering and Technology, Hyderabad is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Engineering in Information Technology.

This is a record of bonafide work carried out by me and the results embodied in this project have not been reproduced or copied from any source. The results embodied in this project report have not been submitted to any other university or institute for the award of any other degree or diploma.

Mr. Mohammed Abdul Rahman (160923737111)

ACKNOWLEDGEMENT

I am very pleased to present this phase-I of our project. This period of my student life has been truly rewarding; a number of people were of immense help to me during the course of research and the preparation of the project.

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Lastly, I acknowledge the support of my faculty members, friends, and family, who inspired and encouraged me throughout this journey.

Mr. Mohammed Abdul Rahman (160923737111)

Vision of the Institute

Lords Institute of Engineering and Technology strives for excellence in professional education through quality, innovation and teamwork and aims to emerge as a premier institute in the state and across the nation.

Mission of the Institute

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To be a frontier in Information Technology focusing on quality professional education, innovation and employability.

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DM1: Adopt innovative teaching learning processes to achieve stated outcomes and sustained performance levels.

DM2: Provide learner centric environment to enrich knowledge, skills and innovation through industry academia collaborations.

DM3: Train the concerned stakeholders on latest technologies to meet the industry needs.

DM4: Inculcate leadership, professional, interpersonal skills and ethical values to address the contemporary societal issues.

Note: DM: Department Mission

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B.E Information Technology Program Educational Objectives (PEOs):

PEO1	Be a competent software engineer in IT and allied industry providing viable solutions.
PEO2	Exhibit professionalism, engage in lifelong learning and adopt to changing industry and societal needs.
PEO3	Be effective as an individual and also cohesive in a group to execute multidisciplinary projects.
PEO4	Pursue higher studies in Information Technology with emphasis on competency and innovation.

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B.E Information Technology Program Outcomes (POs): Engineering

Graduate will be able to:

S.no	Program Outcomes (POs)
1.	**PO1: Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
2.	**PO2: Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
3.	**PO3: Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
4.	**PO4: Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
5.	**PO5: Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
6.	**PO6: The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
7.	**PO7: Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
8.	**PO8: Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
9.	**PO9: Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
10.	**PO10: Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
11.	**PO11: Life-Long Learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

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B.E Information Technology Program Specific Outcomes (PSOs):

PSO1	**Professional Skills:** Identify, analyze requirements, design and implement computer programs in the area of artificial intelligence, machine learning, big data analytics and networking of varying complexity.
PSO2	**Problem-Solving Skills:** Deliver quality service to IT industry and provide solutions to real-life problems applying modern computer languages, software methods and tools.

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Course Outcomes: Project Phase-I
After completion of the project, the student will be able to

S.No	Description	Blooms Taxonomy Level
PP.I.CO1	Demonstrate the ability to synthesize and apply the knowledge and skills acquired in the academic program to the real-world problems.	BTL3
PP.I.CO2	Evaluate different solutions based on economic and technical feasibility	BTL4
PP.I.CO3	Effectively plan a project and confidently perform all aspects of project management.	BTL5
PP.I.CO4	Demonstrate effective written and oral communication skills	BTL4
PP.I.CO5	Prepare the documentation report and perform the presentation of the project work	BTL3

Note: Bloom's Taxonomy Levels

BTL1-Remember	BTL2 – Understand	BTL3 –Apply
BTL4-Analyze	BTL5 –Evaluate	BTL6–Create

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Course Articulation Matrix:

Mapping of Course Outcomes (COs) with Program Outcomes

(POs) and Program Specific Outcomes (PSOs):

Course Outcomes (COs)	Program Outcomes (POs)											Program Specific Outcomes (PSOs)	
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO1	PSO2
PP.I.CO1	3	2	2	-	2	2	-	-	3	-	-	2	2
PP.I.CO2	3	3	2	3	2	-	2	-	3	-	-	-	2
PP.I.CO3	3	3	3	-	3	3	1	2	2	3	2	3	-
PP.I.CO4	3	2	2	3	3	3	3	2	-	-	2	3	3
PP.I.CO5	3	3	3	3	3	3	3	2	3	3	-	3	3
AVG	**3**	**2.6**	**2.5**	**3**	**2.6**	**2.7**	**2.2**	**2**	**2.9**	**3**	**2**	**2.7**	**2.5**

Level:

1- Low correlation (Low), 2- Medium correlation (Medium), 3-High correlation (High)

SDG Mapping:

SDG	Mapped Indicator	SDG	Mapped Indicator	SDG	Mapped Indicator
1 NO POVERTY		7 AFFORDABLE AND CLEAN ENERGY	✓	13 CLIMATE ACTION	
2 ZERO HUNGER		8 DECENT WORK AND ECONOMIC GROWTH		14 LIFE BELOW WATER	
3 GOOD HEALTH AND WELL-BEING		9 INDUSTRY, INNOVATION AND INFRASTRUCTURE	✓	15 LIFE ON LAND	
4 QUALITY EDUCATION		10 REDUCED INEQUALITIES		16 PEACE, JUSTICE AND STRONG INSTITUTIONS	
5 GENDER EQUALITY		11 SUSTAINABLE CITIES AND COMMUNITIES		17 PARTNERSHIPS FOR THE GOALS	
6 CLEAN WATER AND SANITATION		12 RESPONSIBLE CONSUMPTION AND PRODUCTION			

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ABSTRACT

Reliable operation of power transformers is essential for ensuring uninterrupted power supply in industrial and residential sectors. Transformer failures can lead to significant economic losses, reduced system reliability, and increased maintenance costs. To address these challenges, predictive maintenance techniques based on Machine Learning (ML) have gained considerable importance in modern power systems.This project presents a Machine Learning-based Transformer Fault Detection and Remaining Useful Life (RUL) Prediction System aimed at improving transformer reliability and reducing unexpected failures. The proposed framework utilizes Dissolved Gas Analysis (DGA) data containing key transformer gases such as Hydrogen (H_2), Carbon Monoxide (CO), Ethylene (C_2H_4), and Acetylene (C_2H_2). Advanced machine learning algorithms including LightGBM and XGBoost are employed for fault classification and health assessment.The system automatically identifies transformer operating conditions such as Normal, Partial Discharge, Thermal Fault, and Electrical Fault while estimating the Remaining Useful Life of the transformer. Statistical feature extraction techniques are applied to generate meaningful insights from gas concentration data. In addition, Explainable Artificial Intelligence (XAI) using SHAP analysis is incorporated to improve model transparency by identifying the most influential factors affecting predictions.A Django-based web platform provides user authentication, dataset management, prediction services, interactive dashboards, and analytical reports. The proposed system enables early fault detection, accurate health monitoring, and effective maintenance planning, thereby enhancing transformer performance, reliability, and operational efficiency. The combination of Machine Learning, Explainable AI, and web-based deployment offers a scalable and intelligent solution for predictive maintenance in power distribution systems.

Keywords: Smart Grid, Distribution Substation, Predictive Maintenance, Machine Learning, Failure Prediction, Grid Intelligence, Fault Detection, Data Analytics, Equipment Monitoring, Failure Prevention

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